Talon Pilot

Handbook

Includes setup and operation for TalonSync desktop software



TALON OS

Talon Pilot 2.24.0 · TalonSync 1.15.0

Created by NevvyDrone · nevvydrone.com

A virtual handheld for VRChat avatars

Contents

Contents 2

Welcome 5

What's in the box 5

About this handbook 5

Getting to Know Your Talon Pilot 6

Controls 6

The Expressions menu 7

Setting Up 8

What you need 8

Installing 8

The Control Center 8

Anchor points: wings, paws and other hands 9

Updating 9

Check my install 9

What's new 10

Turn on Self Interact 10

Basics 11

Turning Talon Pilot on and off 11

The launcher 12

Tapping 12

Entering text 13

Applications 14

Calendar 14

Contacts 14

Tasks 15

Memos 15

Calc 15

Clock and Stopwatch 16

Prefs 16

TalonSync . . . 17
Expense, Card Info and Security . . . 17
Talon 3D . . . 18
Hypno . . . 18
Compass . . . 19
Scope . . . 19
Hoot Pet . . . 20
TalonCast . . . 21
Owl Oracle . . . 22
Moon & Stars . . . 23
PC Link . . . 24
Laser pointer . . . 24
Holo foil . . . 25
Beam . . . 25
TalonPop . . . 26
Snow Globe . . . 26
Aquarium . . . 27
Terminal and Talon Enhanced BASIC . . . 29
Screensavers . . . 33
Preferences . . . 34
Themes . . . 35
Custom themes . . . 36
Wallpaper . . . 36
Case styles . . . 37
Finishes . . . 37
Your own stylus . . . 38
Desktop Mode . . . 39
Look and click . . . 39
Other ways to tap . . . 39
Size and position . . . 40
Talon Pilot and Other Players . . . 41

Performance . . . 41
Choosing your apps . . . 41
TalonSync Desktop Software . . . 43
Installing on Windows . . . 43
Installing on Linux and macOS . . . 44
Updating . . . 44
First start . . . 44
Synchronizing . . . 45
Your information . . . 46
Files other programs open . . . 47
TalonCast . . . 47
Hypnospiral, Hoot Pet and PDA Remote . . . 48
Now Playing . . . 49
PC Link . . . 50
Settings . . . 51
Linux and macOS notes . . . 52
Where TalonSync keeps your information . . . 52
Questions and Answers . . . 53
Specifications . . . 55
Regulatory and Safety Information . . . 56
Federal Cyberspace Commission notice (fictional) . . . 56
Aerie Spectrum Authority (fictional) . . . 56
Declaration of Conformity (fictional) . . . 57
Safety . . . 57
Battery and disposal . . . 57
Limited warranty (fictional) . . . 57
Trademarks . . . 57

Welcome

Thank you for choosing Talon Pilot, the handheld organizer that lives on your avatar.

Talon Pilot is a pocket-sized personal digital assistant for VRChat avatars, inspired by the handhelds of the early 2000s. It has a color touchscreen, a stylus, four hardware app buttons and a 5-way navigator, and runs **Talon OS**: a calendar, contacts, tasks, memos, a calculator, a world clock, a 3D viewer, a virtual pet, a weather channel and more.

Your Talon Pilot keeps your information with you from world to world, and the companion program **TalonSync** keeps it in step with your computer, the way you would have docked a handheld in its cradle.

What's in the box

Item	What it is
Talon Pilot handheld	The PDA itself, held in one hand (or wing).
Stylus	A pen for your other hand. Its fine tip taps the screen; the wider tip presses the buttons.
Talon OS	The operating system and all built-in applications.
Install Wizard	A Unity tool that puts Talon Pilot on your avatar and updates it later.
TalonSync	Desktop software for Windows, Linux and macOS: sync, backups, weather, music and more.
This handbook and the Quick Start Guide	Everything you need to get going.

About this handbook

Words in this style, such as **Done**, are things you see on the screen. Menu paths are written like **Expressions › Talon Pilot › Power**. Tips and notes appear in shaded boxes.

Talon Pilot is a virtual device: it has no battery to charge, nothing to break and it never runs out of storage. It does, however, appreciate being tapped gently.

Getting to Know Your Talon Pilot

A quick tour of the controls.



Talon Pilot is designed for one-handed holding and two-handed use: hold it in one hand, and tap with the stylus (or a fingertip) in the other.

Controls

Control	What it does
Touchscreen	Tap icons, list rows, buttons and keys.
Home	The little owl Home button in the middle of the screen's bottom bar. Returns to the launcher from anywhere.

Control	What it does
App buttons	Four round buttons that open an app straight away. Out of the box: Calendar, Contacts, Tasks and Memos. You can change them in Prefs › Buttons.
5-way navigator	Up and Down move the highlight. Left goes back. Right and Center (**Ok**) select. Many apps use Left and Right for their own controls.
Stylus	Appears in your other hand whenever the PDA is on, and is put away when you switch off.
Expressions menu	Power, quick app launching, an on-screen 5-way pad and desktop controls. See page-by-page details below.

The Expressions menu

Open your Expressions (Action) menu and choose **Talon Pilot**:

- **Power**: switches the PDA on and off. It chirps while it starts up.
- **Laser** and **Holo foil**: the laser pointer and the case's rainbow shine (if you installed them). Other players see both.
- **Apps**: jump straight to any main app.
- **Controls**: the 5-way navigator and Home as menu buttons, handy if a button is out of reach.
- **Desktop**: controls for VRChat desktop mode (see the chapter *Desktop Mode*).
- **Reset** returns to the launcher and clears the current screen. **Factory Reset** also puts every preference back to its default; your saved text is kept.

Setting Up

Installing Talon Pilot on your avatar takes about a minute.

What you need

- A VRChat avatar project in Unity 2022.3 with the VRChat Avatars SDK 3.
- An avatar with a Humanoid rig (most avatars have one).
- The Talon Pilot package from your purchase.

Installing

1. Unzip the package and copy its **Assets/TalonPilot** folder into your project's **Assets** folder. Unity imports it automatically.
2. Choose **Tools › Talon Pilot › Install Wizard**. On an avatar without Talon Pilot it starts the **Guided Setup**.
3. **Welcome**: type the name you'd like your Talon Pilot to call you.
4. **Avatar**: pick your avatar. The wizard checks it and tells you in plain words if anything needs fixing.
5. **Hand & Tapping**: the hand that holds the PDA, how you tap (Stylus, Finger or Both), the size, and anchor points if you hold things with a wing or paw.
6. **Case**: choose a case style, a finish and its colours, and your own stylus if you like, with a live 3D preview.
7. **Screen**: a starting theme, your own custom themes and a wallpaper picture (all optional).
8. **Extras** and **Review**: the defaults are right for almost everyone. Then press **Install**.
9. **Done**: the self-check runs, the PDA is selected in the Hierarchy. Move and rotate it so it sits nicely in your palm, then upload your avatar.

The Control Center

Open the wizard on an avatar that already has Talon Pilot and you get the **Control Center**: every option in a list on the left in four groups, and a search box. **Device**: Overview, Placement, Case & Colours, Screen. **On the PDA**: PDA Settings, Tasks & Memos, Calendar & Contacts, App Options and Built-in Content, so you can fill in your PDA before you upload. **Setup**: Apps & Features, Desktop, TalonSync & PC,

Performance. **Help**: Support, About. Change anything, then press **Apply to avatar**. Your placement, settings, text and pet are kept. When a newer Talon Pilot is in your project, the Control Center offers **Update**; if pieces are missing it offers **Repair**.

> **TIP**
> If only your text or PDA settings changed, Apply just sends them (it takes a moment). **Revert** undoes anything you haven't applied yet.

Anchor points: wings, paws and other hands

The PDA rides on your hand bone and the stylus on your other hand automatically. If your avatar holds things with a different bone, such as a wing, tell the wizard in the **Anchor points** box (on the Setup page, and on the Avatar page when updating):

1. In the Hierarchy, expand your avatar, then **Armature**, and find the bone. Typing part of its name (for example *wing*) into the Hierarchy's search box helps.
2. Click the bone and press **Use selected**, drag it into the box, or choose it from **Suggestions**.
3. Leave a box empty to use the hand. The wizard shows which bone it will use.

> **IMPORTANT**
> To move the PDA to a different bone later, use Anchor points and **Update**. Don't drag it onto another bone by hand: the PDA's animations find it by its place in the hierarchy, and it would stop switching off and resizing. The wizard warns you if this happens.

Updating

When a new version arrives, copy its **Assets/TalonPilot** folder over the old one and open the wizard: the Control Center shows **Update available**. Press **Update**. Your placement, settings, text and pet are kept. **Tools › Talon Pilot › Patch Avatars to Latest** updates every avatar in the open scene at once.

Check my install

The wizard's Done page, Overview and Support pages have a **Check my install** button. It has Unity's own shader compiler (the one VRChat's upload uses) build all four screen programs, then checks the PDA, its parameters, menu and screen materials, and shows a green tick or a plain explanation for each. If the screen can't compile on your computer, **Use the safe screen for now** swaps in a calm owl-mint

screen with a note instead of magenta. The upload does the same check by itself: an avatar is never uploaded with a pink PDA, and your scene copy isn't touched.

Talon Pilot works alongside VRCFury and Modular Avatar. It is for PC avatars (the screen shader is too big for Quest/Android).

What's new

The first time you switch on a PDA after an update, a small **What's new** card lists the highlights. Tap anywhere or press any button to close it; it won't show again until the next update.

Turn on Self Interact

In VRChat, open **Settings** and make sure **Self Interact** is on. Without it, the screen can't feel your own stylus or finger.

Basics

Turning on, finding your way around and entering text.

Turning Talon Pilot on and off



Starting up

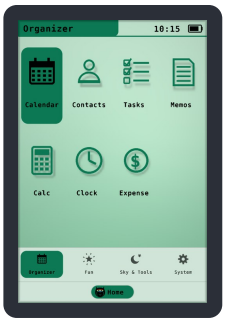
Choose **Expressions › Talon Pilot › Power**. Talon Pilot runs a real power-on self-test: every line is an actual check of your PDA, from its ROM and memory to the clock, touch screen and TalonSync. Then you hear the start-up chirp and the launcher opens.

The test only takes as long as the slowest check, usually under a second with TalonSync running (at most 1.5 seconds). **First boot, set up** appears once on a new avatar. Amber lines are warnings (for example a finger resting on the screen); the PDA still starts.

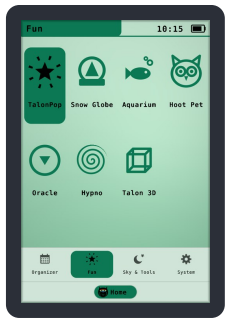
Choose **Power** again to switch off. The PDA and stylus are put away, and everything you were doing is remembered.

Auto-off. In Prefs › Size & Power you can have Talon Pilot switch itself off after 5, 15 or 30 minutes of rest.

The launcher



Organizer



Fun

The launcher sorts your applications into four categories, one per page: **Organizer**, **Fun**, **Sky & Tools** and **System**. The title bar shows the category you are in.

Tap an icon to open it, or move the highlight with the 5-way navigator and press Center. Tap a **tab** along the bottom to change category; Left and Right also carry on into the next category when you reach the end of a row.

Tap the owl Home button on the bottom bar at any time to come back here; you return to the category you left.

Organizer: Calendar, Contacts, Tasks, Memos, Calc, Clock, Expense.
Fun: TalonPop, Snow Globe, Aquarium, Hoot Pet, Owl Oracle, Hypnospiral, Talon 3D.
Sky & Tools: TalonCast, Scope, Compass, Moon & Stars.
System: TalonSync, PC Link, Beam, Terminal, Prefs, Card Info, Security.

Tapping

- Tap with the very tip of the stylus: the screen listens to its fine nib, so it's pinpoint accurate.
- The hardware buttons respond to the whole tip, so they're easy to press.
- Each tap gives the classic tap beep (unless you turn it off in Prefs › Sounds).
- In lists, tap a row to open it; in a detail card, tap anywhere to go back.

Entering text



The keyboard

Wherever you can type (tasks, memos, calendar and contact lines, the Hypno caption), open the item and tap its line, or press Right. The on-screen keyboard appears.

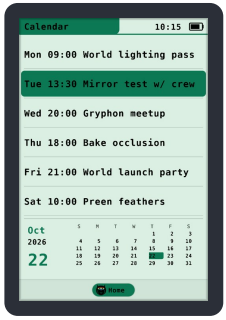
Tap keys to type. **shift** gives capitals, **123** numbers and symbols, and **del** deletes. Each line holds up to 15 characters.

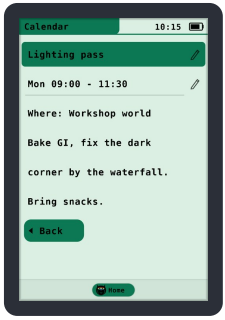
Tap **Done**, or press Center, to save the line. Your text is saved on the avatar and is still there next time you load it.

Applications

Everything that's built into Talon OS.

Calendar





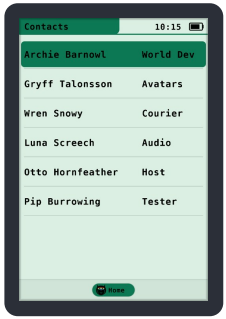
Your upcoming events, soonest first, with a mini month view.

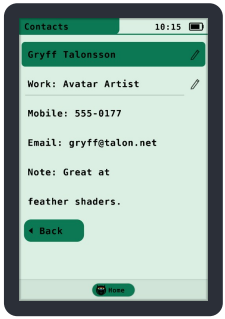
Tap an event (or highlight it and press Center) to see its card: time, place and notes. Tap the card to go back.

If **Edit Calendar & Contacts on the PDA** is on (the default), tap a line on the card to change it with the keyboard.

TalonSync can fill and update your calendar from your computer, including from Outlook, Google and Apple calendars.

Contacts

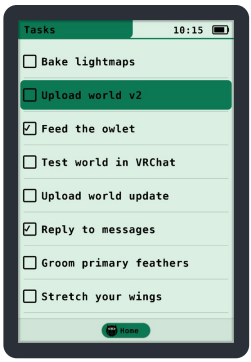




Names with their title or role. Tap a contact for the full card: work, phone, email and a note.

Edit lines on the card just like calendar events, or manage your contacts in TalonSync and sync them across.

Tasks

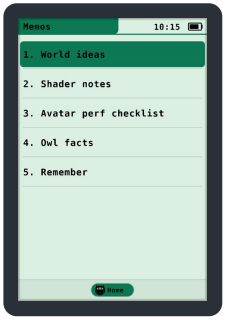


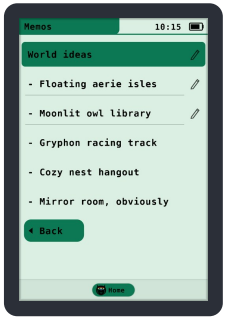
Your to-do list. Tap the box beside a task to tick it off (tap again to un-tick).

Tap the words of a task, or press Right, to type over it.

TalonSync puts your highest-priority open tasks on the PDA and brings your ticks back to your computer.

Memos

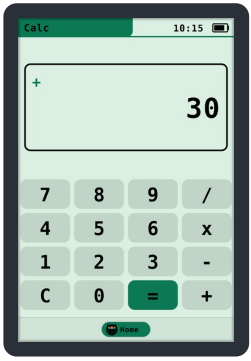




Short notes. The list shows each memo's title; tap one to read it.

Tap any line of a memo to edit it. Write longer memos in TalonSync and they're laid out on the PDA for you.

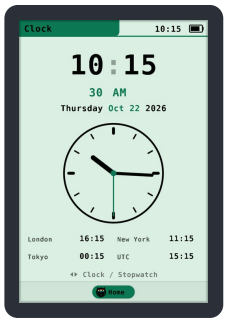
Calc



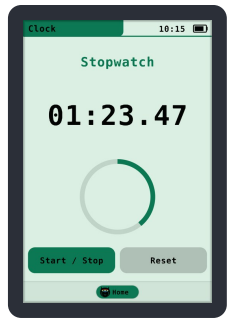
A four-function calculator with a paper tape.

Tap the keys, or use the 5-way navigator: move to a key and press Center. The tape keeps the steps of your calculation.

Clock and Stopwatch



Clock



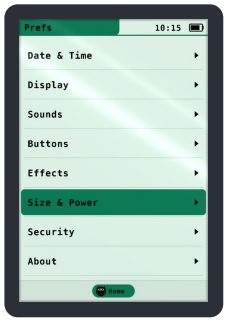
Stopwatch

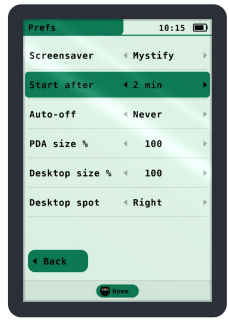
A large clock with the date, an analog face and four world clocks (London, New York, Tokyo and UTC).

Talon Pilot sets itself from VRChat's clock automatically. To set it by hand, choose Prefs › Date & Time › Set time › Manual.

Press Left or Right to switch to the **Stopwatch**, then tap Start / Stop and Reset.

Prefs



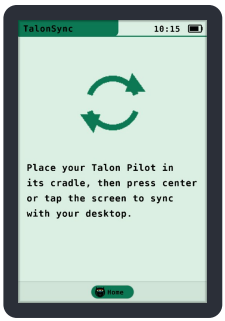


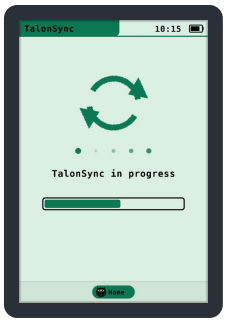
All of Talon Pilot's settings, in eight categories. Tap a category to open it.

Change a setting by tapping the ◂ or ▸ at either end of its row, or highlight it and press Left or Right. Tap **Back** when you're done.

The full list is in the chapter *Preferences*.

TalonSync



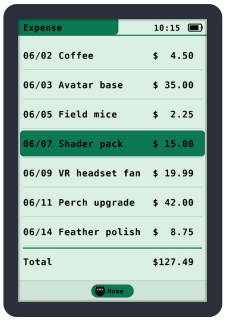


Synchronizes your PDA with the TalonSync program on your computer.

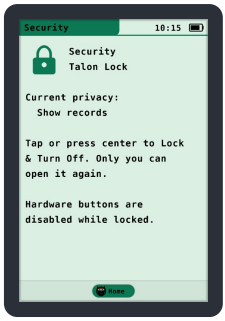
Tap the screen or press Center to start. The arrows spin while each part of your data is exchanged, and the PDA chirps when it's done.

When music is playing on your computer, a Now Playing card appears here too. See the chapter *TalonSync Desktop Software*.

Expense, Card Info and Security



Expense



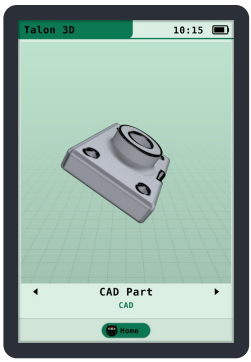
Security

Expense keeps a simple list of purchases with a running total.

Card Info shows your Talon Pilot's storage card, firmware version and owner name.

Security locks your PDA. Tap **Lock** (or press Center): the PDA locks and switches off. When you switch it on again it shows the lock screen; tap the screen to unlock. While locked, the hardware buttons do nothing. Prefs › Security › Lock at power on locks it every time it starts.

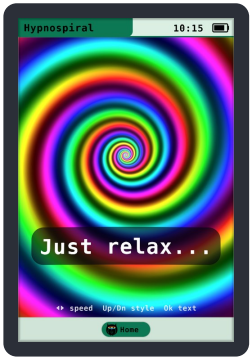
Talon 3D



A little 3D model viewer: a cube frame, a torus, a machined part, an octahedron, a blob and four wireframe models.

Left and Right change the model; Up and Down change the look (Solid, Hologram or CAD); Center starts or stops the spin.

Hypno



A hypnotic spiral with a caption of your choice.

Left and Right change the speed, Up and Down change the style (rainbow spiral, black and white spiral, rainbow rings), and Center lets you type a new caption.

It can also be your screensaver. TalonSync has a Hypnospiral page with a live preview.

Compass



A glass-cockpit navigator, styled like a modern aircraft's flight displays, for exploring worlds. It reads how you hold the PDA and where it is in the world. The display looks like a real matte LCD sitting just behind a cover glass: it shifts behind the bezel as you move your head (real depth in VR), and its lit symbols glow in worlds with bloom.

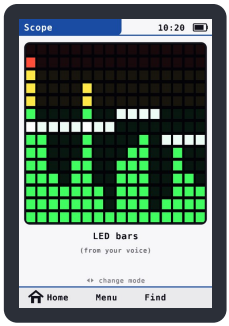
Top: the flight display: an attitude indicator (sky and ground with a pitch ladder and roll scale), your speed on the left tape, your height on the right tape, and your heading. **Bottom:** a moving map that turns with you, with a 50 m range ring, a compass arc, and **HOME** (the world's spawn point, its centre) with a magenta course line, its bearing and distance. The bar along the top shows ground speed, track, the bearing home and pitch.

Scope

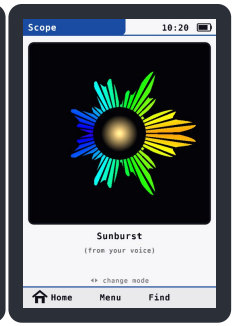


An oscilloscope that reacts to music (in worlds with AudioLink) or your voice. Press Left or Right to switch between nine styles: Spectrum, Waveform, Voice level, **LED bars** with falling peaks, **Mirror bars**, **Sunburst**, **Plasma**, **Warp tunnel** and **Band history**.

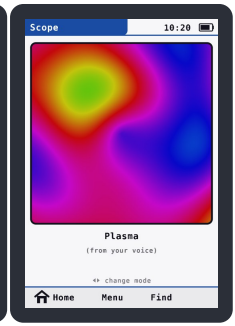
Spectrum and Waveform need AudioLink; the other styles also run from your voice, so they work in any world. Other players see it react too.



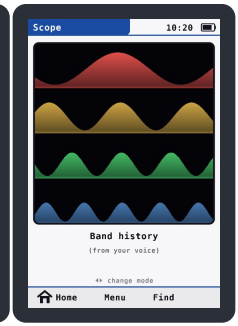
LED bars



Sunburst

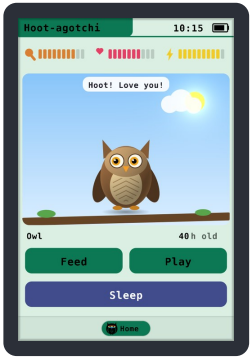


Plasma



Band history

Hoot Pet



Your own owlet to look after. It hatches from an egg and grows over time.

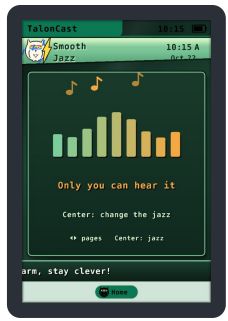
Keep an eye on its food, joy and energy, and tap **Feed**, **Play** or **Sleep** to look after it. A well-cared-for owl is a happy owl.

Your pet keeps living while the PDA is switched off. TalonSync can feed it and play with it from your computer.

TalonCast



Current conditions



Smooth jazz

Your own little weather channel, with real local weather delivered by TalonSync. The pages cycle by themselves: current conditions, the three-day forecast and today's highlights, with a scrolling ticker along the bottom.

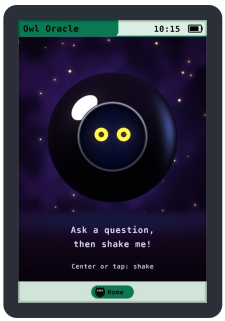
Left and Right flip pages. Center changes the smooth jazz: **Off**, **Only I can hear it**, or **Everyone nearby**.

Other players see your temperature and weather on your PDA too (never your city).

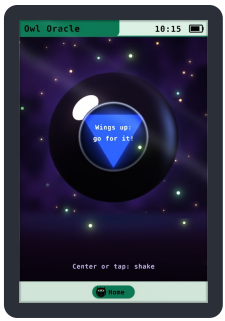


Extended forecast

Owl Oracle



Ask

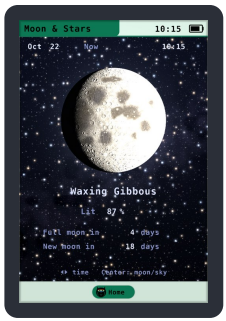


The answer

Ask the oracle a yes-or-no question (out loud, or just in your head), then press Center or tap the screen to shake it.

After a moment of swirling, the answer floats up in the window. Shake again for another opinion. The owls are always honest, if not always clear.

Moon & Stars



Moon



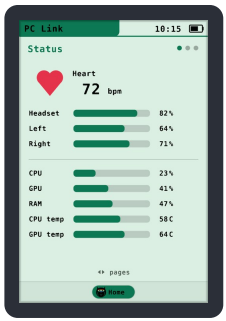
Tonight's sky

Tonight's real moon, worked out from the date and time: its phase, how much of it is lit, and how many days until the next full and new moon. The moon is drawn as a real ball with its true face: the dark seas where they really are, craters that cast shadows along the edge of the light, Tycho's bright rays, and the faint earthshine on its dark side; the sunlit moon and the brightest stars glow in worlds with bloom. It floats just in front of the screen, with the stars and the Milky Way far behind it: in VR each eye sees its own view, so it has real 3D depth.

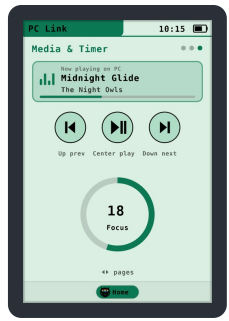
Press Center (or tap the middle) for the star chart: the sky over your head right now, with the brightest stars, the best-known constellations, and the moon and sun. North is at the top and east on the left, as if you held it up over your head.

Left and Right move the time an hour at a time, up to half a day each way. Up and Down change your latitude; TalonSync sets it from your TalonCast city.

PC Link



Status



Media and Timer

Your computer on your PDA, delivered by TalonSync. Left and Right flip between three pages:

Status: your heart rate (from a heart-rate app), headset and controller battery, and your PC's CPU, GPU and memory. The headset's charge also fills the battery icon in the title bar.

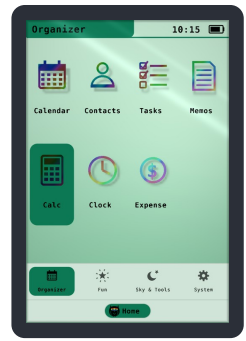
Messages: the latest message from your computer (calendar reminders, Twitch mentions and more). Press Center to clear them.

Media and Timer: what's playing, with Previous, Play/Pause and Next buttons that control the music on your PC (Up, Center, Down do the same), and a focus timer: tap it to start or stop. The PDA chirps when a focus or break ends.

Laser pointer

A red laser shines out of the middle of the case's top edge. Switch it on and off with **Laser** in the Talon Pilot Expressions menu; it works with the PDA switched off too. Other players see it. Where the beam meets a wall, the floor or a person, a bright dot appears (in worlds that provide depth information, which most do; elsewhere you see just the beam). It is a pointer for showing things, so please don't shine it in people's eyes. Leave it out in the Install Wizard (Apps & features) if you don't want it.

Holo foil



Holo foil on the icons

A rainbow shine like a holographic trading card covers the case and the launcher's icons, and shifts as you tilt the PDA or move your head. On the case it follows the world's lighting, so it never glows in a dark room. It starts off: switch it on with **Holo foil** in the Talon Pilot Expressions menu; other players see it. Leave it out in the Install Wizard (Apps & features) if you don't want it.

Beam



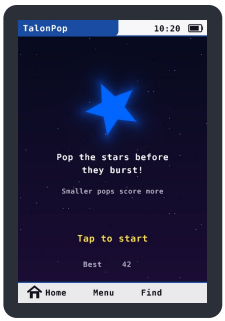
Experimental: Beam doesn't work as intended yet, so treat it as a preview. It may not deliver your message.

Send a short message (up to 15 letters) to a friend's Talon Pilot, the way handhelds used to beam to each other.

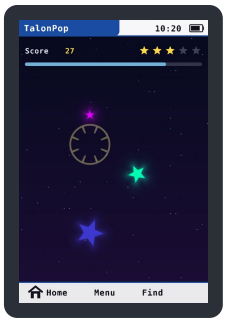
Tap **`Write & beam`** (or press Center), type your message, and press Done. Hold your PDA close to theirs, within about a hand's width, until the beaming animation stops (about seven seconds).

Their PDA chirps and opens Beam with your message. It only works between two Talon Pilots, and only while both are switched on.

TalonPop



Ready



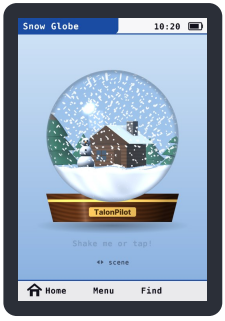
Popping

Stars appear around the screen and grow. Tap each one before it bursts: the smaller you catch it, the more it's worth (3, 2 or 1 points).

A star that bursts costs one of your five lives, and the stars grow faster as the round goes on. A round lasts a minute. Your best score is kept.

Tap the screen or press Center to start, and again to play another round.

Snow Globe



Just shaken



Night

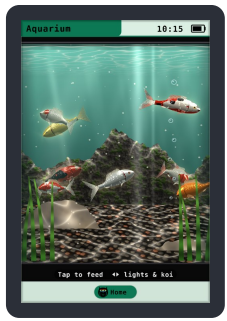
A 3D log cabin, pine trees and a snowman under a water-filled glass ball. Give the PDA a really hard shake (or tap the globe, or press Center) and the snow whirls up, then slowly drifts down and settles.

Left and Right change the scene: day, dusk with the windows lit, or a starry night with smoke from the chimney.

Aquarium



Reef tank



Koi tank

A bowfront tank you look into through the screen. It is drawn from where your eye really is, so in VR each eye sees its own view: the tank has real depth behind the screen, and the curved front glass brings the nearest fish slightly out of it. On a flat screen it shifts as you move.

The **reef tank** has clownfish living in their anemone, a yellow tang, a royal gramma, a flame angel, a copperband butterflyfish and a pair of green chromis among live rock, corals and a sea fan. The **koi tank** has eight koi (kohaku, sanke, showa, tancho and metallic ogon) over river stones and pebbles. Each fish has fins of spines and see-through webbing, gills that open and close as it breathes, a mouth, and shiny scales.

Tap anywhere in the tank to drop food there: the fish swim over and nibble for about ten seconds. Center drops more in the same place.

Left and Right change the lights: day, sunset, the blue night light, and a **blacklight** that makes corals and fish fluoresce. After the four reef lights come the same four for the koi tank.

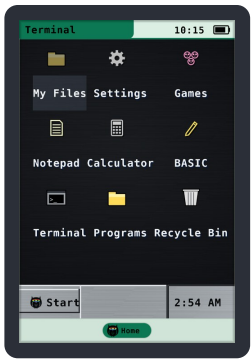


Night light



Blacklight

Terminal and Talon Enhanced BASIC

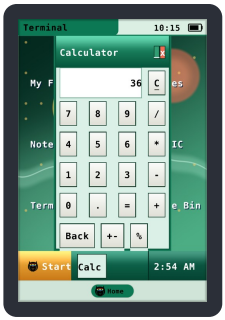


TalonOS Enhanced

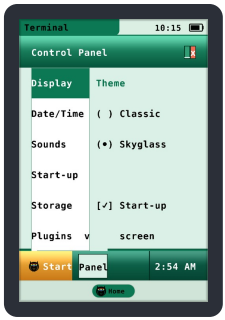
A 30 x 8 green-screen terminal. Your PC is the CPU: Talon Enhanced BASIC runs inside TalonSync, and the PDA is its screen and keyboard. The CPU is an **experimental** TalonSync feature and is off until you opt in: in TalonSync, open **Experimental** in the sidebar and switch on **Terminal CPU**. Until then it says **No CPU found**. The same page shows the Terminal's screen on your PC, so you can type, press the PDA's buttons, click the TalonOS Enhanced desktop, and write BASIC programs in a proper editor, with or without VRChat.

Nothing loads by itself. When the Terminal finds the CPU it asks what to start: **1** TalonOS Enhanced, **2** Talon Enhanced BASIC, or **3** the local terminal, which has its own commands like an old serial terminal (**HELP**, **TIME**, **DATE**, **CPU**, **FILES**, **ECHO**, **LIBRARY**, **BASIC**, **MENU**). **BYE** leaves BASIC for the local terminal. Type **TALONOS** (or **WIN**) anywhere to open the desktop; if you were on the desktop when the PDA switched off, it comes back when you switch it on.

TalonOS Enhanced is a small windowed OS that runs in TalonSync: a desktop with icons, windows you can hide (_) and close (**x**), a taskbar with a button for each open window and a clock, and a **Start** menu with Programs, Accessories, Games, Documents, Settings, Help and Shut Down. **Tap** anything on the screen, or use the buttons: Up and Down move between the controls in a window, Center presses, Left goes back or closes and Right opens Start. Type on the PDA keyboard and press Done: the text goes into the box with the cursor. On the desktop, typing a program's name (NOTEPAD, CALC, GAMES) opens it.



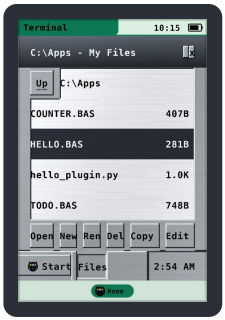
Skyglass



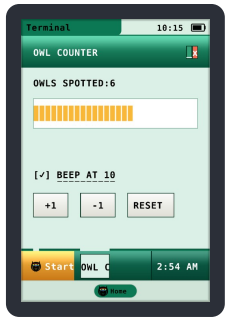
Control Panel

Control Panel (Start › Settings, or the Settings icon) sets the look and how it behaves: **Classic**, brushed aluminium with graphite title bars on a dark desktop, or **Skyglass**, mint glass in the owl colours with a gold Start button over an abstract mint, coral and gold desktop; the start-up screen; a 12 or 24-hour clock and seconds; beeps; and whether the desktop comes back when the PDA switches on. **System** shows your PC's load and TalonSync's uptime. Settings and high scores are kept in **talonos.json** in TalonSync's data folder.

Accessories: Notepad (type lines and press Done; Save keeps your notes on your PC), **Calculator**, **Date/Time** and **System Monitor**. **My Files** shows your notes and BASIC programs: open them, run a program, or delete it to the **Recycle Bin**, where you can restore it or empty the bin.



My Files



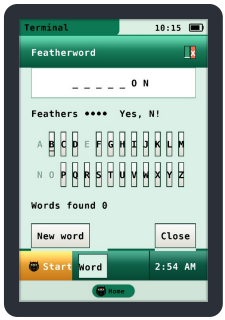
A BASIC app

Files and folders: My Files shows drives and folders like a classic PC: **C:\Documents** for your notes, **C:\Programs** for BASIC programs, **C:\Apps** for installed apps, the **Recycle Bin**, and **A:**, a read-only disk with the built-in BASIC programs. Open, New folder, Rename, Delete, Copy and Paste work as you'd expect; Notepad has Open and Save As boxes. Everything stays inside TalonSync's data folder on your PC.

Write your own apps in BASIC: a program that uses **WINDOW** runs in its own window with buttons, text boxes, check boxes and progress bars, and can read and write files (**OPEN**, **PRINT #**, **INPUT #**). Put it in C:\Apps and it appears in Start › Programs. HELLO, TODO and COUNTER are there to learn from. **Task Manager** shows what's running and can end a stuck app; if an app crashes, only that app closes. Python plugins are possible too, but are off until you allow them in Control Panel. TalonSync's docs folder has the full guide (TalonOS_SDK.md).



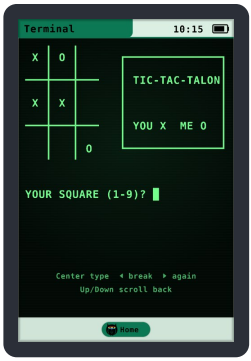
Tic-Tac-Talon



Featherword

Games are real windowed apps you play by tapping: Guess My Number, Rock Paper Talon, Feather Pile, Moon Lander, Tic-Tac-Talon, Featherword, Aerie Keeper, Talon Dice, Find the Mouse, Ponder the Owl to talk to, and Math Drill. Your best scores are kept.

BASIC Programs (the Programs icon) holds the same games written in BASIC, plus a BASIC tour and demos. Choose one and press **Run** (or tap it twice) to run it full screen in BASIC; **List** shows how it is written and **Edit** opens it in BASIC to change it. In text mode, type **LIBRARY**, then a number and **R**. BASIC and the Terminal run full screen too and come back to the desktop when they finish (or press Left). **Shut Down** in the Start menu closes TalonOS.



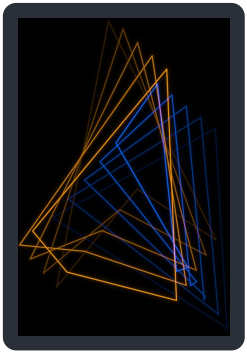
Line graphics

Line graphics: like a serial terminal's line-drawing set, the screen draws real lines that join up across rows. In BASIC, **BOX** row, column, height, width draws a box, **LINE** row1, col1, row2, col2 draws a straight or 45° line, and **LOCATE** row, column moves the cursor (rows 0-7, columns 0-29). Lines that cross join into corners and crossings by themselves. **CHR$(5)** to **CHR$(31)** print shades, half blocks, a ball, diagonals and every box piece.

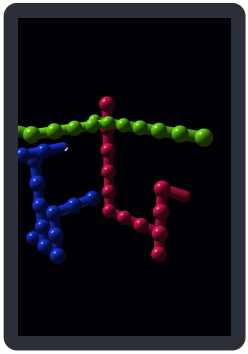
Press Center to type a line, then Done to send it. Lines with a number are stored as a program (**10 PRINT "HI"**); anything else runs at once. **RUN**, **LIST**, **NEW**, **SAVE "NAME"**, **LOAD "NAME"** and **FILES** work like the old home computers; type **HELP** for the rest. Programs are saved on your PC.

Left stops a running program, Right repeats your last line, Up and Down scroll back. A line of exactly 15 letters carries on into the next one, so press Done on an empty line to finish it.

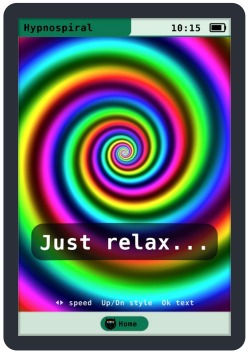
Screensavers



Mystify



3D Pipes



Hypnospiral

After a few minutes of rest Talon Pilot shows a screensaver. Any tap or button press wakes it (without also pressing whatever was underneath). Choose the screensaver and how soon it starts in Prefs › Size & Power.

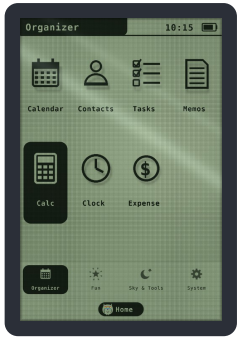
Preferences

Make Talon Pilot your own.

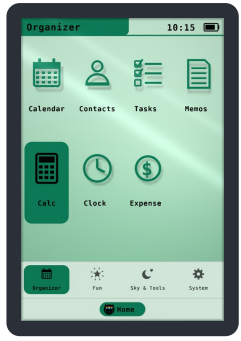
Category	Setting	Choices
Date & Time	Hour, Minute	Used when Set time is Manual
	Time format	12-hour, 24-hour
	Set time	Automatic (VRChat's clock), Manual
Display	Brightness	10 to 100%
	Theme	Classic LCD, Mint Owl (the default), Owl Night, Feather Gray, Gryphon Gold
	Wallpaper	Plain, Aurora, Morning mist (the default)
	Screen effect	None, LCD grid, Scanlines
Sounds	System sound	Off, On
	Volume	Low, Medium, High
	Tap beep / Game sound	Off, On
Buttons	Button 1 to 4	Any of the main apps
Effects	3D depth	On, Off. The depth you see in VR: everything sits a little behind the glass, and the Aquarium, Moon & Stars, Snow Globe, Owl Oracle, Hoot Pet and more have real 3D. Turn it off if it bothers you.
	Glitter	Off, On. Tiny multicoloured flakes float at three depths and flash as they catch the light while you move your head.
	Glass glints	Off, On. Light streaks slide across the screen's glass as you move, like a phone catching a lamp.
	Touch ripples	Off, On. A soft mint ring spreads from wherever you tap the screen.
	Idle glow	Off, On. After 20 seconds untouched, the screen's edge breathes slowly in mint.

Category	Setting	Choices
	Floating motes	Off, On. Marine snow in the Aquarium, fireflies around the Owl Oracle.
Size & Power	Screensaver	Off, Mystify, 3D Pipes, Hypnospiral
	Start after	1, 2, 5 or 10 minutes
	Auto-off	Never, 5, 15 or 30 minutes
	PDA size %	60 to 160 (VR). Other players see your size too.
	Desktop size %	50 to 200 (desktop mode only)
	Desktop spot	Right, Center, Left (desktop mode only)
Security	Lock at power on	Off, On
About		Version and credits

Themes



Classic LCD



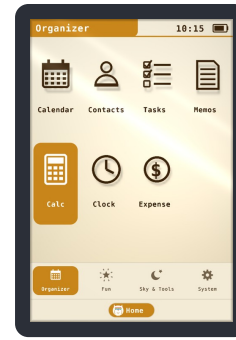
Mint Owl



Owl Night



Feather Gray

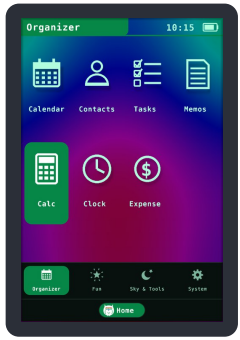


Gryphon Gold

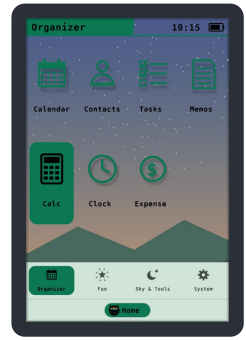
Other players see your PDA in your theme too.

Custom themes

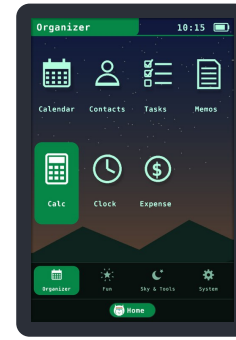
Make up to three of your own in the Install Wizard (Screen › Custom themes): five colours (background, text, highlight, highlight text, buttons), a selection style and a name. They appear after the built-in five in Prefs › Display › Theme, and other players see them too. Themes are small *.talontheme.json* files you can export and share; the *Assets/TalonPilot/Themes* folder has starting points and a short guide.



A custom theme



Your own wallpaper

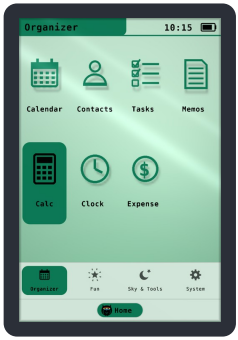


Both together

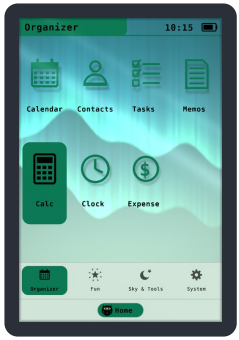
Wallpaper

Prefs › Display › Wallpaper chooses the Home screen's wallpaper: **Plain**, **Aurora** (curtains of mint and teal light rippling across the sky at two depths) or **Morning mist** (the default: soft mist drifting over pale mint, in your theme's own colours on the other themes). Choose your own picture in the Install Wizard (Screen › Wallpaper): it is cropped to the screen's shape, resized and compressed

automatically, and a Dim slider keeps icons readable. Use pictures you have the rights to.



Morning mist



Aurora



Aurora on Owl Night

Case styles

Classic, Slim Metal, Rugged Field, Retro Brick or Clear Plastic, each with your own case and button colours (Install Wizard › Case & Colours). Every style keeps the screen and buttons in the same place, so everything works the same. The Talon Pilot owl is printed on the back.

Finishes



Any case style can have a **finish**: Woodgrain, Brushed Metal, Carbon Fibre, Leather or Speckled retro plastic (Case & Colours › Finish). Picking one suggests a colour; the Case colour then tints the finish, so a pale colour gives light oak and a dark one walnut. The owl logo is printed on every finish. A finish adds one small texture and nothing else.

Your own stylus

Drag your own stylus model (any model or prefab from your Project) into **Case & Colours › Stylus**. Talon Pilot fits it into the stylus's place: the same length, the tip where the old tip was, the same spot in your hand, so nothing needs moving. The narrower end becomes the tip; tick **The tip is at the other end** if it guessed wrong, and use **Turn it** to spin it (to put a clip on top, say). It keeps its own materials. Keep it under 20,000 triangles; the Performance page counts it. **Use the Talon Pilot stylus** switches back.

Desktop Mode

Using Talon Pilot in VRChat without a headset.

When you play VRChat on a desktop, Talon Pilot notices automatically and moves itself in front of your view (low and to the right, so the Action Menu and the crosshair stay clear). Only you see it there; everyone else still sees it in your hand.

Look and click

1. Start **TalonSync** on your PC (version 1.7 or newer). It gives you a click button.
2. Walk around as usual: the PDA follows your view.
3. Stand still for a moment and it **parks** in the world.
4. Turn your view with the mouse so the dot in the middle of your screen is over a button or key.
5. Press the **middle mouse button** to tap it. Hold it to keep pressing.

The PDA comes back in front of you when you walk off, when you turn more than about 35 degrees away from it, and when you sit down. **Expressions › Talon Pilot › Desktop › Bring PDA in front** brings it back any time.

Other ways to tap

- **Choose another button** in TalonSync › Settings › Desktop mode: a mouse side button or an F-key. You can also give **Bring PDA in front** a button there. The click button only works while the VRChat window is in front.
- **Expressions › Talon Pilot › Desktop › Tap where I look** does the same thing without TalonSync: aim with the dot, open the menu and press it.
- **Fist gesture** (an option in the Install Wizard): a right-hand Fist (Right Shift + F2) taps; switch back to Idle (Right Shift + F1) to let go.
- **Cursor joystick** (an option in the Install Wizard): moves an on-screen cursor; **Tap at cursor** taps there. It's off by default because it costs a little frame rate, even in VR.

Size and position

Prefs › Size & Power › **Desktop size %** and **Desktop spot** (right, center or left) change the desktop PDA only. Your VR size is kept separately.

> **NOTE**
> Parking uses VRChat's own constraints (Avatars SDK 3.7 or newer). With an older SDK the PDA simply stays in front of you.

Talon Pilot and Other Players

What people around you see.

Others see	Details
The app you have open	Every app, switched within about a quarter of a second.
Your theme and PDA size	Exactly as you've set them.
The Scope	Reacting to your real voice.
The Compass	With your real movement speed.
TalonCast	Your temperature, weather icon and day or night. Never your city, forecast or ticker.
Smooth jazz	Only if you choose Everyone nearby.
Beam	Only while you're beaming: a few invisible sensors on your PDA carry the message to a Talon Pilot held next to yours.

Your tasks, memos, calendar, contacts and pet stay private: other players see the apps, but not your information.

> **NOTE**
> All of this travels on a single shared value, so Talon Pilot uses at most **11 synced bits** of your avatar's budget: 8 for the shared value, 1 for power, 1 that simply says “I'm beaming” and 1 for TalonCast's shared jazz. Leave out Beam and TalonCast and it's 9.

Performance

Talon Pilot is built to be light. When it's switched off, almost all of its work stops; while it's on, each app only does the work for what's on screen. Players further away see a simplified screen that costs very little to draw.

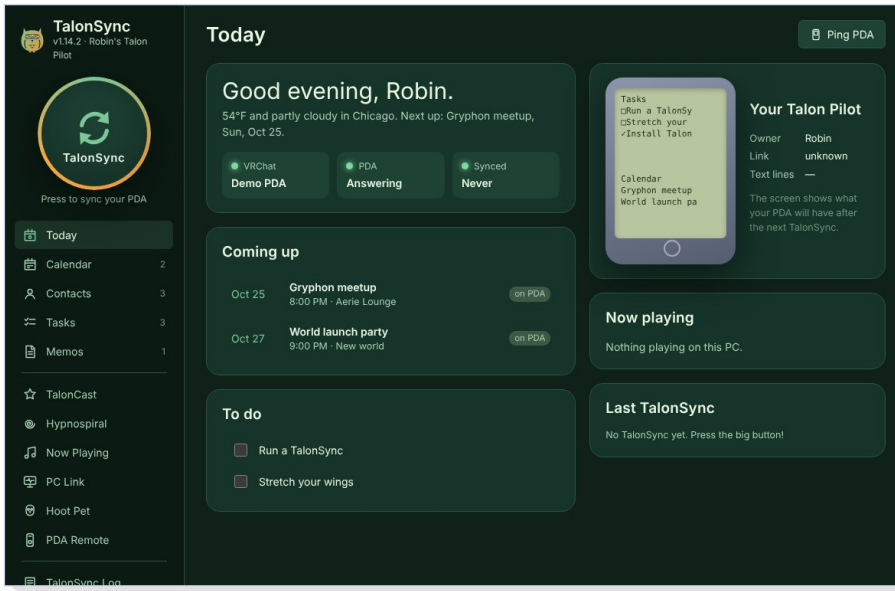
Choosing your apps

Control Center › Apps & Features › **Apps on your PDA** lists the apps you can leave out: Talon 3D, Hypnospiral, Compass, Scope, Hoot-agotchi, TalonCast, Owl Oracle, Moon & Stars, PC Link, Beam, TalonPop, Snow Globe, Aquarium and Terminal. Each shows what it costs. PC Link starts unticked: tick it if you use TalonSync's PC stats

and messages. An app you leave out takes its parameters, animator states, layers and objects with it, and the launcher closes the gap. Leaving out Hoot-agotchi saves two animator layers (it also resets your pet if you add it back later); TalonCast and PC Link save the most parameters; Beam saves a synced bit and 16 contacts. **Lightest** leaves out all fourteen; **All apps** puts them back. The everyday apps are always there.

TalonSync Desktop Software

Keep your Talon Pilot and your computer in step.



TalonSync's Today page

TalonSync is the companion program for your computer. It synchronizes your calendar, contacts, tasks, memos and settings with your PDA, keeps backups, reads and writes the file formats other programs use, puts what you're listening to on your PDA, and runs TalonCast, the weather channel. It talks to VRChat over OSC, VRChat's built-in way for programs on your computer to talk to your avatar.

Installing on Windows

1. Install Python 3.13 or 3.14 from python.org. During setup, tick **Add python.exe to PATH**.
2. Unzip TalonSync and double-click **Build TalonSync.bat**. It makes a private environment, installs what it needs, tests itself and builds **TalonSync.exe**.
3. Run **TalonSync.exe**. It lives in the system tray (the owl icon), so closing the window keeps it running.

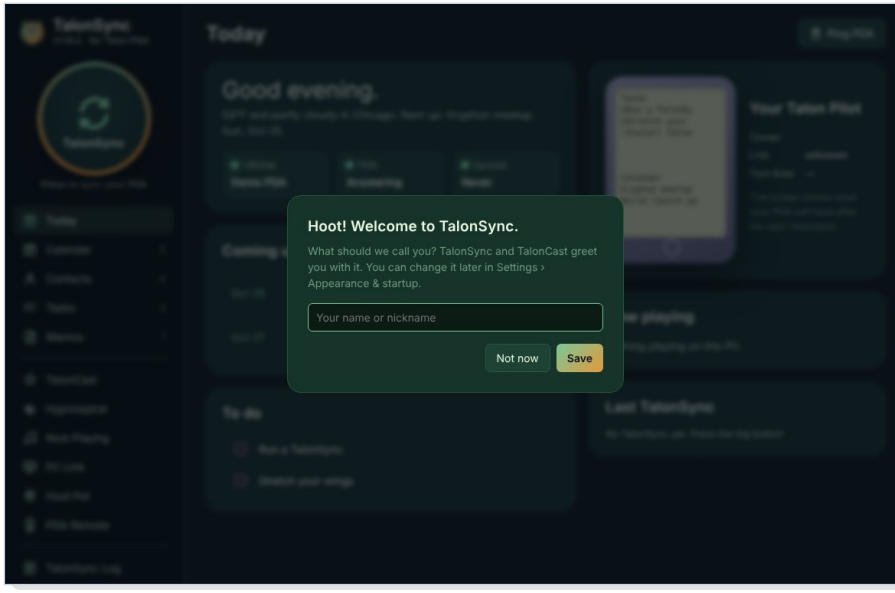
Installing on Linux and macOS

- **Linux:** in a terminal, go to the TalonSync folder and run `./Run\ TalonSync.sh`.
- **macOS:** double-click **Run TalonSync.command**. The first time, macOS may say it's from an unidentified developer: right-click it, choose **Open**, then **Open** again.
- The first run sets everything up and tests itself, which takes about a minute. After that it just starts.

Updating

The Windows program has its files built in, so after unzipping a new version: quit TalonSync (right-click the tray owl, then Quit), unzip over the old folder, and run **Build TalonSync.bat** again. The version number is shown under the TalonSync name. If the program is older than the files beside it, a yellow banner tells you to rebuild.

First start



TalonSync asks what to call you

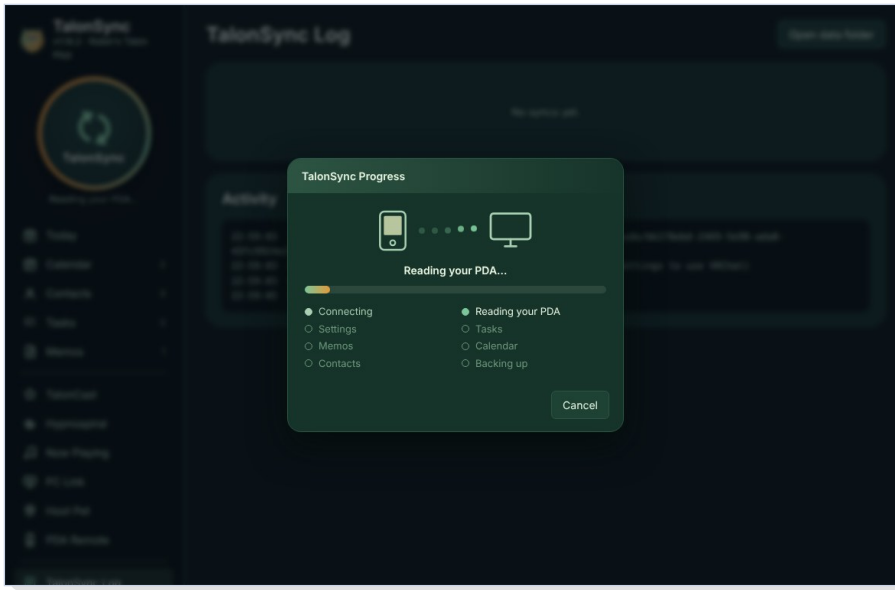
TalonSync asks what to call you. Then:

1. In VRChat, open your Action Menu › **Options** › **OSC** and choose **Enabled**.
2. Wear your avatar with Talon Pilot (installed with the TalonSync PC link, which is on by default).

3 The two lights at the bottom of TalonSync's sidebar turn green: VRChat connected, and Talon Pilot answering.

> **IMPORTANT**
> If you used this avatar with OSC before installing Talon Pilot, VRChat keeps an old OSC list for it. Open Settings › Connection › **Connection doctor**, press **Fix**, then switch avatars once.

Synchronizing



The TalonSync Progress window

Press the big **TalonSync** button (or Ctrl+H), or tap TalonSync on your PDA. Your PDA jumps to its sync screen and the arrows spin, while the progress window on your computer shows each step. The PDA chirps when it's finished, and a backup is saved automatically.

Conduits

Each kind of information has a conduit, set in Settings › TalonSync:

Setting	What happens
Synchronize the files	Changes on either side are merged (the default).
Desktop overwrites handheld	Your computer's copy replaces the PDA's.

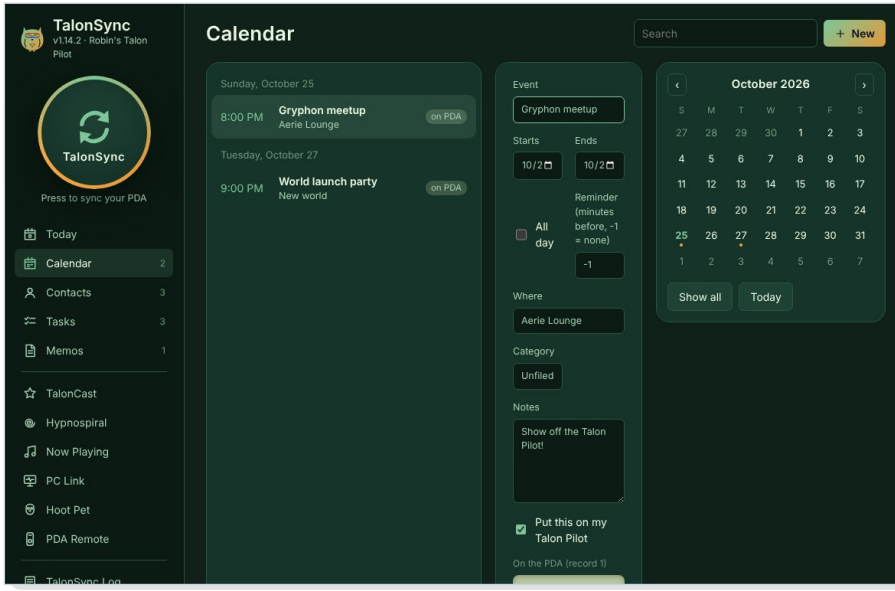
Setting	What happens
Handheld overwrites desktop	The PDA's copy replaces your computer's.
Do nothing	This kind of information is left alone.

The conduits are: Handheld settings, Tasks, Memos, Calendar, Contacts and the Hypno caption. When the same thing has changed on both sides, TalonSync keeps the PC version, the PDA version or both, whichever you choose.

Live backup

TalonSync keeps a copy of everything your PDA remembers (typed lines, task ticks, settings, the Hoot Pet, best scores) the moment it changes, so switching worlds never loses anything. When your avatar loads, TalonSync compares the PDA with the copy; if the PDA came up empty, it offers **Put it all back** (or does it by itself if you switch on **Put a wiped PDA back automatically** in Settings › Files & backups). The Backups page lists the last few snapshots, and you can restore any of them. Nothing leaves your computer.

Your information



Calendar

Calendar, Contacts, Tasks and Memos each have their own page for adding and editing. Tick **Put this on my Talon Pilot** to choose what goes to the PDA; the preview shows exactly how it will look there.

Files other programs open

	Export	Import
Contacts	vCard (.vcf), CSV	vCard 2.1 / 3.0 / 4.0, Outlook or Google CSV
Calendar	iCalendar (.ics), CSV	.ics from Outlook, Google, Apple; CSV
Tasks	.ics, Markdown checklist, CSV	The same
Memos	Markdown, CSV	Markdown, text, CSV
Everything	TalonSync backup	Unity TalonSync JSON

Drag files onto **Import & Export** to bring them in; duplicates are skipped. You can also set a folder that receives every format after each sync, and a watch folder that imports anything you drop into it.

TalonCast

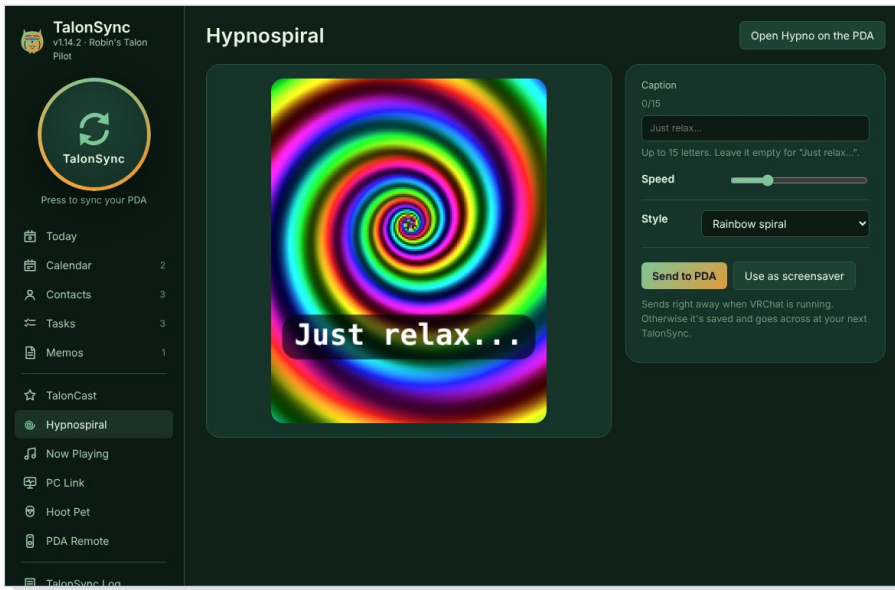


TalonCast on your computer

The weather channel, full size on your computer, with smooth jazz. Type your city in the **Location** panel and press **Find**, or press **Use where this PC is**. Choose units, pages, look, the ticker and the jazz volume; TalonSync sends the weather to your PDA's TalonCast too.

- **Ticker on the PDA**: a weather summary, your own message (up to 45 letters) or off.
- **Jazz on the PDA**: Off, Only I can hear it, or Everyone nearby can hear it.
- Weather comes from Open-Meteo: free, with no account.

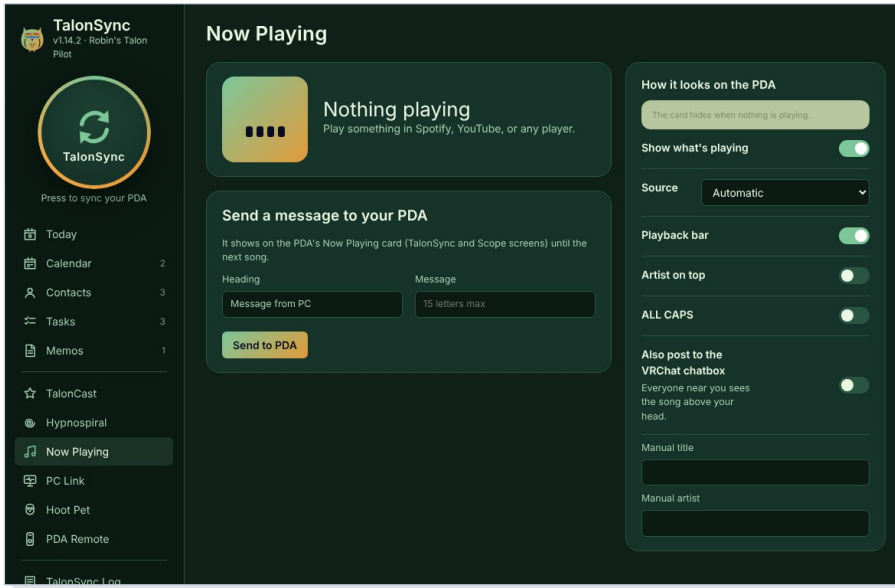
Hypnospiral, Hoot Pet and PDA Remote



Hypnospiral

- **Hypnospiral**: a live preview, with caption, speed and style. **Send to PDA**, **Open Hypno on the PDA**, or **Use as screensaver**.
- **Hoot Pet**: see how your owl is doing, and feed it or play with it from your computer.
- **PDA Remote**: open any app on your PDA, and switch it on or off.

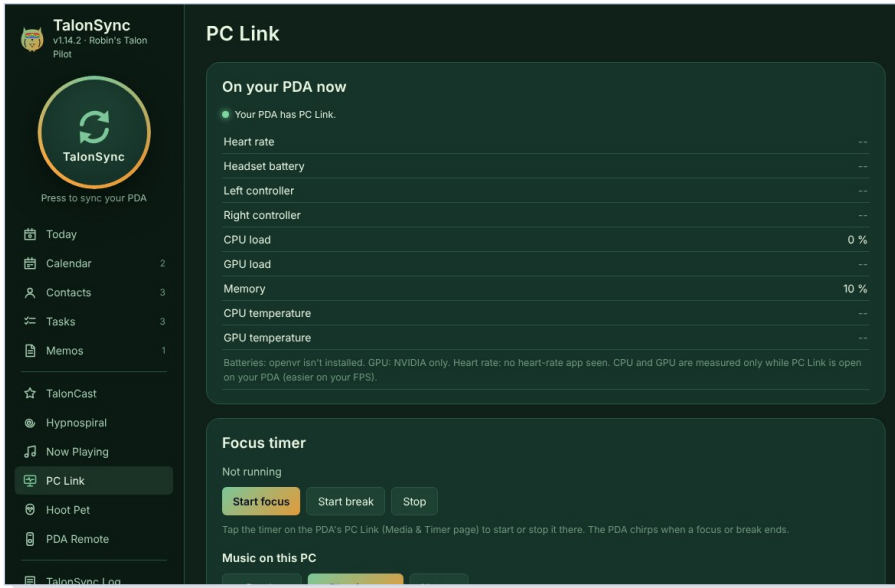
Now Playing



Now Playing

TalonSync shows the song playing on your computer on the PDA's Now Playing card (on the TalonSync and Scope screens), with a progress bar. It reads your computer's media controls: Spotify, YouTube in a browser, Apple Music, VLC and most other players. It can also post the song to your VRChat chatbox, and you can send your PDA a short message from this page.

PC Link



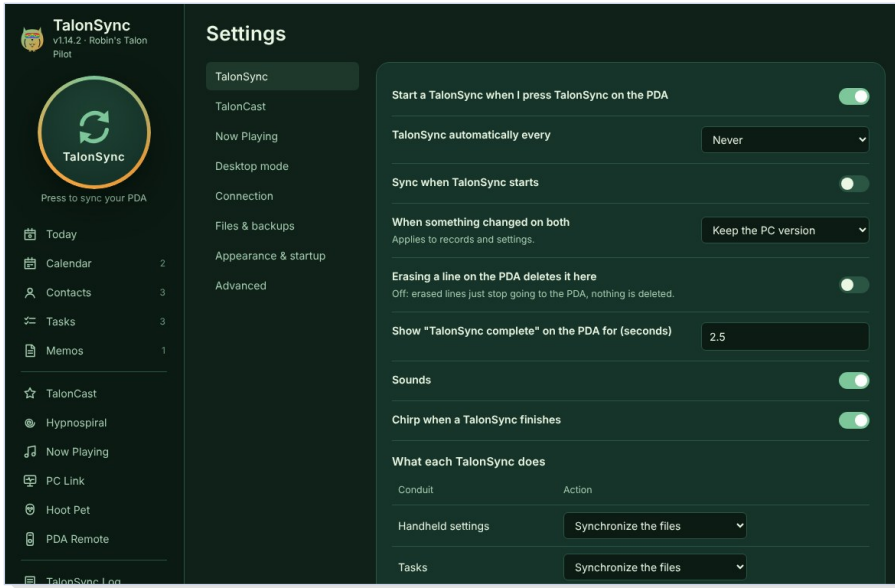
PC Link

TalonSync feeds the PDA's PC Link app. Each part can be switched on or off on this page:

- **CPU, GPU and memory**: load and temperatures. GPU figures need an NVIDIA card; CPU temperature is available on Linux.
- **Headset and controller battery**: from SteamVR, while it's running.
- **Heart rate**: from any heart-rate app that sends your heart rate to VRChat as the avatar parameter **HR** (most popular ones can). The wizard adds that parameter for you.
- **Messages**: type one here, or let calendar reminders and Twitch @mentions arrive on their own. On Linux your desktop notifications come through too. Other programs can send one over OSC: `/talonpilot/message "title" "text"` to TalonSync's port.
- **Music**: the PDA's buttons press your computer's media keys (Windows), control your music player (Linux) or Spotify and Music (macOS).
- **Focus timer**: focus and break lengths, and whether it keeps cycling.
- **Twitch chat**: enter your channel to show the chat on the TalonCast ticker (choose Twitch chat under Ticker on the PDA), or to turn @mentions into PDA messages. No login is needed.

> **NOTE**
> Windows doesn't let ordinary programs read other apps' notifications, so on Windows, messages come from TalonSync itself: reminders, Twitch, the Send box and OSC.

Settings



Settings

Tab	Highlights
TalonSync	Answer the PDA, automatic syncing, sync on start, conflicts, conduits, sounds.
TalonCast	Location, units, pages, look, animation, old-TV glass, ticker, clock, jazz.
Now Playing	Source, playback bar, line order, ALL CAPS, chatbox.
Connection	Addresses and ports, OSCQuery, the Demo PDA, the connection doctor.
Files & backups	How many backups to keep, automatic export, the watch folder.
Appearance & startup	Themes, accent color, your name, start when you log in, start in the tray, close to the tray.
Advanced	Bus timing and retries, for busy instances.

> **TIP**
> Try everything without VRChat: Settings › Connection › **Demo PDA**.

Linux and macOS notes

- **Linux with Steam Proton:** VRChat's OSC works just as on Windows. If TalonSync keeps waiting for VRChat, turn **OSCQuery** off in Settings › Connection, then switch avatars once.
- **Now Playing on Linux** reads your desktop's media controls (playerctl or gdbus). On macOS it reads Spotify and Apple Music.
- **macOS** has no tray icon, so closing the window quits TalonSync.
- **VRChat on another computer:** on the Mac, set Listen on to 0.0.0.0 and VRChat address to the PC's IP address, turn OSCQuery off, and start VRChat with the launch option `--osc=9000:MAC-IP:9001`.

Where TalonSync keeps your information

Windows: %APPDATA%\TalonSync. For a portable copy, put an empty file called *portable.txt* beside the program. A dated backup is made after every sync (Import & Export › Backups).

Questions and Answers

If something isn't working the way you expect.

The screen doesn't respond to my taps.

Turn on **Self Interact** in VRChat's settings. Tap with the very tip of the stylus. If you chose Finger, use the index finger of your other hand.

The stylus doesn't disappear when I switch off.

The PDA or stylus was probably dragged onto another bone after it was installed. Open the wizard and press **Update**; it reconnects them and keeps them where you put them.

The PDA is in the wrong place.

Select it in the Hierarchy and move or rotate it. To attach it to a wing or paw, use **Anchor points**.

Unity says the shader compiler timed out.

Choose **Tools › Talon Pilot › Allow Long Shader Compiles**, restart Unity and Unity Hub, then right-click Assets/TalonPilot and choose Reimport. The first compile can take several minutes.

The PDA screen is pink or magenta.

Open the Install Wizard and press **Check my install**. It lists what's wrong in plain words. **Use the safe screen for now** shows an owl-mint screen until it's fixed, and the upload swaps it in by itself if needed.

Unity says a script "already contains a definition".

Two copies of Talon Pilot are in the project (often an old folder renamed or moved). Open the Install Wizard: it moves the stray copies to the trash. Or delete the older folder yourself, keeping Assets/TalonPilot.

My PDA forgot everything after I switched worlds.

Start TalonSync before VRChat. Its live backup keeps a copy of the PDA all the time and offers **Put it all back** when the PDA comes up empty.

TalonSync says it's waiting for VRChat.

Check that OSC is enabled in VRChat (Action Menu › Options › OSC). Run Settings › Connection › Connection doctor › Fix, then switch avatars once. On Linux, try turning OSCQuery off.

TalonSync is missing features I've read about.

You're probably running an older TalonSync.exe. Quit it from the tray and run Build TalonSync.bat again. The version is shown under the TalonSync name.

TalonCast says No weather yet.

Start TalonSync on your computer and choose your city on its TalonCast page.

My frame rate is lower than I'd like.

Switch the PDA off when you're not using it: almost all of its work stops. Leave out apps you don't use (Control Center › Apps & Features › Apps on your PDA). If you don't use VRChat on a desktop, turning off Desktop mode support (Control Center › Desktop) removes the desktop parts. Control Center › Performance shows the budget.

On desktop I can't tap anything.

Start TalonSync 1.7 or newer and press the **middle mouse button** while the dot is over the PDA (VRChat must be the window in front). Or use Expressions › Talon Pilot › Desktop › **Tap where I look**. If the PDA drifted off to the side, use **Bring PDA in front**.

I need help.

Open **Talon_Pilot_Help.html** in the Documentation folder (or the wizard's Support › **Help page**) for quick fixes. Still stuck? Press **Copy support info** at the bottom of the Install Wizard (or Tools › Talon Pilot › Copy Support Info), and in TalonSync › Settings › Connection. Paste both into your help request and email it to **nevvydrone@nevvydrone.com**: they hold the versions, your options, the self-checks and recent errors, and nothing personal.

I want to start over.

Factory Reset (Expressions › Talon Pilot) puts every preference back. To remove Talon Pilot completely, use Control Center › Support › Remove.

Specifications

Operating system	TalonOS "Nightwing" (Prefs › About shows the version)
Processor	Animator state machine
Graphics	TalonOS Core shader, three programs (OS, Play, FX)
Display	320 × 480 virtual pixels, color, touch-sensitive
Memory	48 saved text lines of 15 characters
Input	Touchscreen, 5-way navigator, 4 app buttons, stylus, Expressions menu
Sound	Tap beep, start-up chirp, smooth jazz
Network	9 to 11 synced bits, depending on the apps you keep
Connectivity	TalonSync over OSC with OSCQuery
Size	78 × 121 mm at 1×; 60 to 160% in game
Requirements	Unity 2022.3, VRChat Avatars SDK 3, Humanoid avatar. TalonSync: Python 3.10.1 or newer; Windows 10/11, Linux or macOS
Battery	Not required

Regulatory and Safety Information

The fine print, for a device made entirely of light.

> **NOTE**
> This chapter is a lighthearted tribute to the regulatory pages that came with handhelds of the era. Talon Pilot is a virtual device; nothing here is an actual statement by, or compliance claim with, any government agency or standards body.

Federal Cyberspace Commission notice (fictional)

This virtual device complies with Part 15¾ of the Federal Cyberspace Commission Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful lag, and (2) this device must accept any interference received, including interference from avatars with more PhysBones than polygons.

This equipment has been tested and found to comply with the limits for a Class Hoot digital device. These limits are designed to provide reasonable protection against harmful frame drops in a residential instance. This equipment generates, uses and can radiate smooth jazz and, if not installed and used in accordance with this handbook, may cause harmful interference to nearby conversations. If this equipment does cause such interference, which can be determined by turning the jazz off and on, the user is encouraged to try one or more of the following measures:

- Reorient or relocate your avatar.
- Increase the separation between your PDA and other players' ears.
- Choose Only I can hear it in TalonCast.
- Consult your instance owner or an experienced world builder for help.

Changes or modifications to the shader not expressly approved by you, the person who installed it, could void your ability to compile it.

Aerie Spectrum Authority (fictional)

This Class Hoot digital apparatus complies with the Aerie Spectrum Authority's rules for interference-causing equipment. Cet appareil numérique de la classe Hibou est conforme à la réglementation de l'Autorité du Spectre de l'Aire.

Declaration of Conformity (fictional)

Trade name	Talon Pilot
Model	TP-5.5
Responsible party	You, the owl holding it
Conforms to	Part 15¾ (Class Hoot), Aerie Spectrum Authority ASA-003 (Hoot)

Safety

- **Use in moving worlds:** do not read memos while falling off a map.
- **Stylus:** do not attempt to tap other players' screens with your stylus without asking first.
- **Hypnospiral:** if you or anyone in your instance is sensitive to flashing or moving patterns, use the Black and White style at a low speed, or avoid the app. Look away and take a break if you feel unwell.
- **Volume:** keep game sounds and smooth jazz at comfortable levels.
- **Pets:** Hoot Pet is a virtual owl. No real owls were neglected in the making of this device.
- **Heat:** Talon Pilot does not get warm. Your GPU might.

Battery and disposal

Talon Pilot contains no batteries, no heavy metals and no recyclable parts. When it reaches the end of its life, simply remove it with the Install Wizard. Please do not throw your avatar in the trash.

Limited warranty (fictional)

Talon Pilot is warranted against defects in pixels and workmanship for as long as the owl icon remains on your screen. This warranty does not cover damage caused by dragging the PDA onto another bone by hand, uploading in the middle of a shader compile, or feeding your Hoot Pet after midnight. Your actual rights as a purchaser are those described where you bought Talon Pilot.

Trademarks

Talon Pilot, Talon OS, TalonSync, TalonCast, Hoot Pet and Owl Oracle are names of this product. VRChat is a trademark of its owner; Talon Pilot is an independent product and is not affiliated with or endorsed by VRChat. Other names are the property of their respective owners.

Talon Pilot 2.24.0 · TalonSync 1.15.0. Created by NevvyDrone · nevvydrone.com. Screens shown are from the TalonOS Core renderer; your colors and content may differ.